

CONVEX CONCENTRATION FORCES THE HOFFMANN–JORGENSEN ORLICZ CONDITION

AGENT LANE 06

STATUS

This packet gives a full answer to Remark 4.9 of Adamczak–Kutek, *On Orlicz spaces satisfying the Hoffmann–Jorgensen inequality* [1]. They prove that condition (HJ) implies their convex-Lipschitz concentration inequality (4.9), and ask whether (HJ) is also necessary when $\Psi(x) \leq Ce^{Cx}$. It is necessary.

SOURCE QUESTION

The source defines $\Psi(x) = e^{\psi(x)} - 1$ and says that Ψ satisfies (HJ) if there is $K > 0$ such that, for all large s, u ,

$$\psi(su) \leq K(s \log(1 + s) + s\psi(u)).$$

For $\Phi(t) = \Psi(t^2)$, Proposition 4.7 proves a concentration inequality for convex 1-Lipschitz functions under (HJ). Immediately after it, Remark 4.9 states:

We note that unlike in the case of inequalities presented in Section 4.1 we do not know if under an additional assumption that $\Psi(x) \leq Ce^{Cx}$, the condition (HJ) is necessary for the inequality (4.9) to hold.

for any convex and 1-Lipschitz function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ and $t > 0$, where $c_\Phi \in (0, \infty)$ is a constant depending only on Φ .

Remark 4.8. It is easy to show that in the above theorem one can replace the mean by a median, as under (4.9) the median and the mean are apart by at most $C_\Psi \|\max_{i \leq N} |X_i|\|_\Phi$, so it is enough to adjust constants.

Remark 4.9. We note that unlike in the case of inequalities presented in Section 4.1 we do not know if under an additional assumption that $\Psi(x) \leq Ce^{Cx}$, the condition (HJ) is necessary for the inequality (4.9) to hold.

4.3. Orlicz boundedness of partial sums of random series with independent summands. Let us now pass to an application related to integrability of partial sums of series of independent random variables in Banach spaces, which was the original motivation of Hoffmann-Jørgensen. Recall that a sequence $(Y_n)_{n \geq 1}$ of random variables with values in $(F, \|\cdot\|)$ is said to be stochastically bounded, if for every $\varepsilon > 0$ there exists $K > 0$ such that for all $n \geq 1$, $\mathbb{P}(\|Y_n\| \geq K) < \varepsilon$. In [15] Hoffmann-Jørgensen proved the following result (see Theorem 3.1 and Corollary 3.2 therein).

Theorem 4.10. *Let $X_n, n \geq 1$ be independent random variables with values in a Banach space $(F, \|\cdot\|)$ and let $S_n = X_1 + \dots + X_n$. If the sequence $(S_n)_{n \geq 1}$ is stochastically bounded, then the following conditions are equivalent:*

- (i) $(S_n)_n$ is bounded in L_p ,
- (ii) $\sup_n \|X_n\| \in L_p$,
- (iii) $\sup_n \|S_n\| \in L_\infty$.

THEOREM

Let Ψ be an Orlicz function, write $\Psi = e^\psi - 1$, and set $\Phi(t) = \Psi(t^2)$. Assume that $\Psi(x) \leq Ae^{Ax}$ for all large x , for some $A < \infty$. Suppose that there are constants $a, A_0 > 0$ such that for every N , every random vector $X = (X_1, \dots, X_N)$ with independent real coordinates, every convex 1-Lipschitz $f : \mathbb{R}^N \rightarrow \mathbb{R}$, and every $t > 0$,

$$(1) \quad \mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq A_0 \exp\left(-a \frac{t^2}{(\mathbb{E} \max_{i \leq N} |X_i|)^2}\right) + \frac{A_0}{\Phi(at / \|\max_{i \leq N} |X_i|\|_\Phi) + 1}.$$

Then Ψ satisfies (HJ).

IDEA OF THE PROOF

Use the same sparse Bernoulli variables as in the source's necessity proof for the empirical-process inequalities, but feed them to the convex function $x \mapsto \|x\|_2$. If there are at least s rare hits, the Euclidean norm is of size $\sqrt{s}$. Meanwhile the maximum coordinate has expectation of order $1/\Psi(u)$ and Φ -norm of order $u^{-1/2}$. Therefore the second term in (1), evaluated at $t \simeq \sqrt{s}$, is of order $\exp[-\psi(csu)]$. The rare-hit event has Poisson probability at least $\exp[-C(s \log(1+s) + s\psi(u))]$. Comparing the upper and lower bounds forces the (HJ) inequality.

PROOF

We shall use constants $C, c \in (0, \infty)$ depending only on the constants in (1) and on Ψ . Their values may change from line to line.

Fix u large enough that $\Psi(u) > 2$, and fix $s \geq 1$. For each large N , let $p_N = (N\Psi(u))^{-1}$, let $Y_{N,1}, \dots, Y_{N,N}$ be independent Bernoulli variables with $\mathbb{P}(Y_{N,i} = 1) = p_N$, and set

$$X_{N,i} = Y_{N,i} - p_N, \quad X^{(N)} = (X_{N,1}, \dots, X_{N,N}).$$

Let $K_N = \sum_{i=1}^N Y_{N,i}$, and let Z be a Poisson random variable with parameter $\lambda = 1/\Psi(u)$. Then K_N converges in total variation to Z .

Take the convex 1-Lipschitz function $f_N(x) = \|x\|_2$. If $K_N \geq \lceil 16s \rceil$ and N is so large that $p_N \leq 1/2$, then

$$f_N(X^{(N)}) \geq (1 - p_N)\sqrt{K_N} \geq 2\sqrt{s}.$$

Also

$$\mathbb{E}f_N(X^{(N)}) \leq (\mathbb{E}\|X^{(N)}\|_2^2)^{1/2} = (Np_N(1 - p_N))^{1/2} \leq \Psi(u)^{-1/2} \leq 1.$$

Thus, for $s \geq 1$,

$$\{K_N \geq \lceil 16s \rceil\} \subset \{f_N(X^{(N)}) - \mathbb{E}f_N(X^{(N)}) \geq \sqrt{s}\}.$$

Letting $N \rightarrow \infty$, Stirling's lower bound gives

$$(2) \quad \liminf_{N \rightarrow \infty} \mathbb{P}(f_N(X^{(N)}) - \mathbb{E}f_N(X^{(N)}) \geq \sqrt{s}) \geq \exp\{-C(s \log(1 + s) + s\psi(u))\}.$$

Indeed, the single term $\mathbb{P}(Z = \lceil 16s \rceil) = e^{-\lambda} \lambda^{\lceil 16s \rceil} / \lceil 16s \rceil!$ has this size, since $\log \Psi(u) \leq \psi(u)$.

We now estimate the two parameters in (1). Let

$$W_N = \max_{i \leq N} |X_{N,i}|.$$

If no Bernoulli success occurs, then $W_N = p_N$; otherwise $W_N \leq 1$. Hence

$$\mathbb{E}W_N \leq \mathbb{P}(K_N \geq 1) + p_N \leq Np_N + p_N \leq \frac{2}{\Psi(u)}$$

for N large.

For the Φ -norm, put $B_N = \mathbf{1}_{\{K_N \geq 1\}}$. Since $\mathbb{P}(B_N = 1) \leq 1/\Psi(u)$,

$$\mathbb{E}\Phi(\sqrt{u} B_N) = \mathbb{P}(B_N = 1)\Psi(u) \leq 1,$$

and so $\|B_N\|_\Phi \leq u^{-1/2}$. Since $W_N \leq B_N + p_N$ and $\|\mathbf{1}\|_\Phi$ is a fixed finite constant, taking N still larger if necessary yields

$$(3) \quad \mathbb{E}W_N \leq \frac{2}{\Psi(u)}, \quad \|W_N\|_\Phi \leq \frac{C}{\sqrt{u}}.$$

Applying (1) to f_N with $t = \sqrt{s}$, using (3), and letting $N \rightarrow \infty$, we obtain

$$(4) \quad \limsup_{N \rightarrow \infty} \mathbb{P}(f_N(X^{(N)}) - \mathbb{E}f_N(X^{(N)}) \geq \sqrt{s}) \leq C \exp\{-c s \Psi(u)^2\} + C \exp\{-\psi(csu)\}.$$

Here the second term uses $\Phi(r) = \Psi(r^2)$:

$$\Phi(a\sqrt{s}/\|W_N\|_\Phi) \geq \Psi(csu).$$

Combining (2) and (4), and writing

$$R(s, u) = C(s \log(1 + s) + s\psi(u)),$$

gives

$$(5) \quad e^{-R(s, u)} \leq C e^{-cs\Psi(u)^2} + C e^{-\psi(csu)}$$

for all large s, u .

We extract (HJ) from this comparison. If $cs\Psi(u)^2 \geq R(s, u) + \log(2C)$, then the first term in (5) is at most $e^{-R(s, u)}/2$, and hence

$$\psi(csu) \leq R(s, u) + \log(2C) \leq C(s \log(1 + s) + s\psi(u)).$$

It remains to consider the complementary case

$$cs\Psi(u)^2 < R(s, u) + \log(2C).$$

After increasing the constants and taking s, u large, this implies

$$\Psi(u)^2 \leq C(\log(1 + s) + \psi(u)).$$

Since an Orlicz function is convex, increasing, and nonzero, $\Psi(u) \geq cu$ for all large u . Therefore

$$u \leq C(\log(1 + s) + \psi(u)).$$

Using the assumed upper exponential growth of Ψ , equivalently $\psi(x) \leq Cx + C$ for all large x , we get

$$\psi(csu) \leq Csu + C \leq C(s \log(1 + s) + s\psi(u)).$$

Thus in every case

$$\psi(csu) \leq C(s \log(1 + s) + s\psi(u))$$

for all sufficiently large s, u . Replacing s by s/c and adjusting the constant gives the standard (HJ) condition:

$$\psi(su) \leq K(s \log(1 + s) + s\psi(u))$$

for all large s, u . This proves the theorem.

VERIFICATION AND NOVELTY NOTES

The proof uses only the source paper's definitions, the sparse Bernoulli/Poisson test-array idea already used in its proof of Proposition 4.5, and elementary estimates on Orlicz norms. The new observation is to test the convex concentration inequality with the Euclidean norm.

Local indexes for `fa_banach_001` were searched for arXiv:2310.04163, Hoffmann–Jorgensen, Orlicz, and convex concentration; no duplicate packet or attempt was found. A bounded web search on 2026-06-29 for the exact arXiv id and for the phrases “convex-unbounded”, “Hoffmann–Jorgensen Orlicz condition”, and “necessity convex functions” found only the source paper and no later exact resolution.

HUMAN REVIEW RECOMMENDATION

Review the constants in the final case split, especially the passage from the comparison inequality (5) to the use of the exponential growth assumption. The main probabilistic test and the Orlicz scaling appear robust.

REFERENCES

- [1] R. Adamczak and D. Kutek, *On Orlicz spaces satisfying the Hoffmann–Jorgensen inequality*, arXiv:2310.04163.